Describing Distributions & Linear Regression

Section 1: Interpreting the Least-Squares Regression Line (LSRL)

1 A regression model predicting used car price in dollars from age in years is $\hat{y} = 24000 - 1800x$. Interpret the slope -1800 in context.

a. For each additional year of car age, the predicted price decreases by 1,800 dollars

b. A car loses exactly 1,800 dollars every year

c. When the car is 0 years old, its predicted price is $-1,800$ dollars

d. The correlation between age and price is -1800

2 In the model $\hat{y} = 45 + 3.2x$, what is the predicted value of y when $x = 10$?

a. 77.0

b. 48.2

c. 32.0

d. 145.0

3 Why is the line called the 'least-squares' regression line?

a. It minimizes the sum of the squared vertical residuals $\sum(y - \hat{y})^2$

b. It minimizes the sum of squared distances to the origin

c. It maximizes the square of the correlation coefficient

d. It guarantees all residuals are perfect squares

Section 2: Residuals and Residual Plots

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4 An observed data point is $(x, y) = (5, 38)$. The regression equation is $\hat{y} = 12 + 6x$. What is the residual for this observation?

a. -4

b. 4

c. 42

d. -2

5 What does a positive residual signify?

a. The actual observed response was greater than the model predicted

b. The actual observed response was less than the model predicted

c. The regression line has a positive slope

d. The correlation coefficient must be positive

6 Which pattern in a residual plot indicates that a linear model is appropriate?

a. A random scatter of points with no discernible pattern around the residual $= 0$ line

b. A clear curved U-shape pattern

c. A fan shape where residuals expand as x increases

d. Points clustered strictly along an inclined line

Section 3: Correlation Coefficient r and Coefficient of Determination r^2

7 If the coefficient of determination is $r^2 = 0.64$ and the slope of the regression line is negative, what is the correlation coefficient r?

a. -0.80

b. 0.80

c. -0.64

d. -0.4096

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8 What is the correct statistical interpretation of $r^2 = 0.75$?

a. 75% of the variation in the response variable y is explained by the linear relationship with x

b. 75% of the data points lie directly on the regression line

c. The slope of the regression line is 0.75

d. There is a 75% probability that x causes y

9 Which statement about the correlation coefficient r is FALSE?

a. If the units of x are converted from inches to centimeters, r will increase by a factor of 2.54

b. r is always between -1 and 1 inclusive

c. Switching the roles of x and y does not change r

d. r measures only the strength and direction of linear relationships

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Section 4: Outliers, Leverage, and Influential Points

10 A point that has an extreme x-value relative to the rest of the data is said to have:

a. High leverage

b. A large residual

c. High correlation

d. Negative slope

11 What makes a data point an 'influential point' in linear regression?

a. Removing the point substantially alters the slope, intercept, or correlation

b. It has a residual equal to 0

c. It lies directly at $(\bar{x}, \bar{y})$

d. It has both positive x and positive y coordinates

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12 Does a strong correlation between two variables x and y imply that changes in x cause changes in y?

a. No, correlation does not imply causation due to potential confounding variables

b. Yes, as long as $|r| > 0.95$

c. Yes, provided the relationship is strictly linear

d. No, unless the sample size is at least 1,000

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